Fabio Orsi
3D Tech Artist & Developer

fabioorsi.github.io

BRAZIL
+55 19 9 8118 0604
osb.fabio@gmail.com

Profile

As an **Industrial Designer** with a foundation in **Control & Automation Engineering**, I have always worked at the intersections between **creative and technical** disciplines. Focusing on **3D models** and **coding** around them, most of my professional contributions fit the definition of a **Technical Artist**.

Beyond **manual authoring**, I have designed dozens of **automated workflows** that have **scaled the production and management** of hundreds of thousands of **3D assets** for applications in various fields, including **medical**, **manufacturing**, and **product visualization**.

Skills

- **3D MODELING & TEXTURING** — **Creating 3D assets** for **offline rendering**, **real-time visualization** (Web, XR, Game Engines), and **digital manufacturing** (CNC and 3D printing) based on concept art, photo references and technical drawings.
  Blender | Substance Painter | Substance Sampler | GIMP | Inkscape
- **TECHNICAL ART** — **Developing standalone scripts** and **API-integrated plugins** (e.g. for Blender, Substance Painter, GIMP) to **standardize and automate** pipelines for **authoring, processing and managing 3D assets**.
  Blender API | Substance Painter API | GIMP API | Python | Web (HTML/CSS/JS) | Command Line Interfaces (CLI)
- **SYSTEMS INTEGRATION** — **Designing middleware** to connect **enterprise tools** (Odoo, Redmine), **web services** (Google Cloud, AWS, Discord), **3D authoring software** (Blender, Substance), and **game engines** (Unity, Unreal).
  Python | Javascript | C/C++ | C# | Command Line Interfaces (CLI)
- **XR DEVELOPMENT** — **Creating XR apps** for the **Meta Quest** and **Pico** headsets (VR/MR) with integrations to **cloud-based web services** (Google Cloud/Firebase) for **database and 3D asset repositories**.
  Unity | Unreal Engine | Meta Quest SDK | Pico SDK | Javascript | C/C++ | C#
- **PRODUCT DESIGN** — **Engineering** and **designing physical products** at different stages of the development cycle, from **prototyping** to **mass production**, utilizing specialized **CAD software**.
  Autodesk Fusion | Autodesk Inventor | PTC Onshape
- **INDUSTRIAL AUTOMATION** — **Automating digital manufacturing pipelines** (CNC and 3D printing), including tools for **bulk order placement**, **mesh processing**, **production nesting**, and **logistics**.
  Python | C/C++ | Command Line Interfaces (CLI) | G-Code | C#

Experience

3D Artist / Project Manager / Technical Lead
SPASE INC. - San Jose, California, United States (Remote)
Full-time - Jun 2019 - Dec 2025 (6 years and 7 months)

- **Authoring 3D assets** from reference photos and technical drawings for **real-time product visualization** in **e-commerce** and **interior design apps**.
- **Quoting projects** for clients, **directing and briefing a team of diverse artists**, performing final **quality control** and **technical adjustments** on 3D assets.
- **Creating internal tools** to **standardize** and **optimize** production **pipelines** (e.g. plugins for Blender, Substance Painter, Chrome, GIMP).
- **Designing middleware** to **connect** infrastructure (**Google Sheets, Firebase, AWS**), team tools (**Discord**) and 3D software (**Blender, Substance Painter**).

Production Consultant
СМОЛЕНСК МЕБЕЛЬ - Smolensk, Smolensk, Russia (On-Site)
Full-time Freelance - Sep 2018 - Dec 2018 (4 months)

- **Deploying an ERP** at a large-scale factory, **integrating systems** such as CCTV, factory access control, inventory management and CNC machines.
- **Updating technical drawings** and migrating manufacturing data from different sources into the ERP's centralized **production management system**.
- **Developing real-time scheduling algorithms** to synchronize production cycles with **live data** from sales, inventory, and logistics.

3D Artist / Software Developer
DENTAL DIRECT BRASIL - São Paulo, São Paulo, Brazil (On-Site / Remote)
Full-time - Apr 2016 - Sep 2018 (2 years and 4 months)

- **Developing systems** for the **secure acquisition**, **storage**, and **management** of confidential patient information and **3D medical imaging (CT/3D scans)**.
- **Automating 3D medical images manipulation** via custom algorithms for **volumetric analysis**, **parametric mesh generation** and **mesh pre-processing**.
- **Automating the operation** of Drop-on-Powder **3D printers**, optimizing production based on **real-time logistics**, **urgency**, and **nesting efficiency**.

Intern Mechanical Engineer
SHEMPO - Campinas, São Paulo, Brazil (On-Site)
Part-time - Mar 2011 - May 2011 (3-month internship)

- **Creating CAD technical drawings** for parts used in LED signage, using different **production methods** (e.g. injection, casting, sheetmetal, machining).

Education

- **Centro Universitário Belas Artes de São Paulo (FEBASP)**
  Product Design (Industrial Design)
  Graduated
  2014 - 2017
- **Universidade Estadual de Campinas (UNICAMP)**
  Control & Automation Engineering (Mechatronics)
  Incomplete
  2009 - 2013

Fabio Orsi
3D Tech Artist & Developer
fabioorsi.github.io
+55 19 9 8118 0604
osb.fabio@gmail.com

DEVELOPMENT

Proficiencies & Experience

Programming Languages

Python	★★★★★
C / C++	★★★
C#	★★
Web (JS/HTML/CSS)	★★
Java	★

APIs & Environments

Blender API	★★★★★
Substance Painter API	★★★
Google Cloud / AWS	★★★
Meta Quest / Pico SDKs	★
Unity / Unreal Engine	★

TECHNICAL ART & PIPELINE AUTOMATION

My professional experience includes the development of **custom algorithms** to assist in the **production and management** of **3D assets** in **high-volumes**. These tools **automate processes** that are **essential or repetitive**, ensuring **quality and technical consistency** across projects, **increasing productivity**, and **reducing manual overhead**.

Utilizing **APIs** from software like **Blender** and **Substance Painter**, my contributions have optimized the production of 3D assets for **various applications** over the years, automating from quick **ad hoc tasks** to **entire workflows**.

KEY IMPLEMENTATIONS :

Automated & Parametric Asset Production

.mesh_generation	Automatic mesh generation based on prefabs, variable parameters, and non-mesh 3D formats (e.g. DICOM, CAD).
.mesh_processing	Scripts for mesh manipulation on the vertex level based on predefined rules and variable parameters.
.topology_analysis	Tools for topological analysis utilizing general vertex data along ray-casting and gaussian measuring algorithms.
.procedural_textures	Smart Materials and scripts for automatic texture manipulation based on IDs, UV maps and topology features.
.materials_setup	Scripts to automatically set material parameters and assign texture files based on naming and folder hierarchies.
.custom_shaders	Algorithms to generate custom pre-compiled shaders and perform real-time shader adjustments.

High-Volume Asset Management

.bulk_import_export	Plugins to streamline importing and exporting large amounts of assets across different software.
.bulk_optimizations	Algorithms for automatic mesh and texture optimizations based on variable parameters and performance metrics.
.gallery_generation	Tools to automatically generate 3D galleries for quick visualization and inspection of asset packs.

Workflow Standardization

.asset_standards	Tools to inspect and enforce standards for units, texel density, default transforms, texture resolutions and more.
.naming_conventions	Scripts to inspect assets and directories to enforce folder hierarchy and naming conventions.
.environment_sync	Presets and plugins to sync lighting and camera settings across different software and render engines.

XR RESEARCH & DEVELOPMENT *

As an enthusiast of XR technologies, I see a lot of potential in **mixing** the **virtual and real worlds**. Following this interest, I have **prototyped** a few simple **VR/AR applications** to experiment with the implementation of **cloud-based asset repositories**, utilizing **Unity and Unreal Engine** to build for the **Quest and Pico headsets**.

These apps consist of **simple UIs** in which users can preview and select from a list of **3D assets**, which are then **dynamically fetched** from a remote **Firebase server** and loaded into the XR environment, allowing users to **view and manipulate** them in space (in VR or MR passthrough).

KEY REMARKS :

Research Goals

.game_engines	Experiment with the main game engines and learn about their editors, programming languages and render pipelines.
.xr_development	Learn more about developing for XR, using the Meta Quest and Pico SDKs for Unity and Unreal Engine.
.cloud_repositories	Try solutions for real-time access to high volumes of 3D assets utilizing cloud-based repositories in XR applications.

Development Challenges

.quest_pico_firebase	Google-imposed API limitations for Firebase in the Quest and Pico ecosystems required REST workarounds.
.ar_shadow_casting	Blending virtual shadows in AR required custom shaders and fine tuning to achieve performance and visual quality.

* While these remain small **personal R&D initiatives**, they provided **critical insights** into development cycles for popular game engines and the **infrastructure** required to **deploy cloud-connected spatial computing apps**.

DEVELOPMENT
Proficiencies & Experience

• SYSTEMS INTEGRATION & INDUSTRIAL AUTOMATION

A significant part of my work experience includes the development of **middleware** to connect **internal and 3rd-party systems** (e.g. **ERPs/CRMs**, **cloud-based web services**, **production software**, **CNC machines**), assisting in the **flow of information** between clients, team members, departments and companies.

These **integration layers** include connections between **Redmine**, **Odoo**, **AWS**, **Google Cloud**, **Discord**, manufacturing machines (**CNCs/3D printers**) and production software (**Blender**, **Substance Painter**).

KEY IMPLEMENTATIONS :

Digital Manufacturing

.cad_cam_database	Tools to manage manufacturing data in local and cloud-based storage systems (e.g. STLs and other CAD/CAM files).
.mesh_pre_processing	Automatic mesh pre-processing for CNC machines and 3D printers, triggered by scheduling or real-time events .
.quality_control	Tools to assist and automate quality control procedures in digital files and manufactured products.
.automatic_nesting	Algorithms to group objects for production with different optimization goals (e.g. time reduction, waste reduction, etc).
.runtime_scheduling	Algorithms to generate custom pre-compiled shaders and perform real-time shader adjustments.

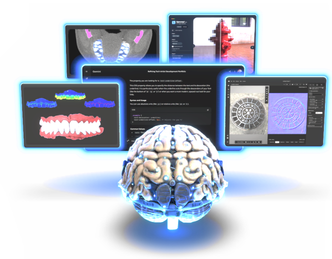
Healthcare Data Management & Processing

.patient_database	Connecting clinics, cloud databases and medical software, with tools such as anonymization and encryption.
.medical_imaging	Algorithms for ML-based segmentation and classification, automatic mesh generation, and more.

Cloud Operations

.erp_integration	Centralizing internal and 3rd-party systems into ERPs and CRMs for efficient management and collaboration.
.database_sync	Routines to sync separate databases and enable real-time updates across systems, projects and companies.
.remote_teams	Bots and plugins connecting IM platforms (e.g. Slack, Discord) to cloud services and production software.

• MACHINE LEARNING & ARTIFICIAL INTELLIGENCE



While over-reliance in AI risks creative stagnation and job obsolescence, its **strategic integration** can act as an **enabler and enhancer** for artists, developers and other professionals across all fields.

My personal experience with **ML/AI** started in 2016 when developing **computer vision algorithms** for the medical field, and extends to today with uses that include **texture creation**, referencing **technical documentation** for APIs, quick **debugging**, and more.

KEY UTILIZATIONS & IMPLEMENTATIONS :

Computer Vision & Medical Imaging

.2d_computer_vision	Implementing ML-based computer vision algorithms for automatic segmentation and classification in DICOM layers.
.3d_spatial_vision	Developing CNNs for segmentation and classification in 3D meshes based on topology measurements and vertex data.
.ai_3d_generation	Utilizing information from the 2D and 3D computer visions to guide the generation of 3D meshes and materials.

Textures & PBR Materials

.upscaling	Image upscaling for textures and reference photos via CNN algorithms.
.pbr_materials	Using AI tools to estimate PBR parameters for materials and generate textures based on reference photos.

LLMs & Software Development

.documentation	Quick ad hoc implementations and referencing technical documentation for APIs/SDKs.
.debugging	Quickly finding bugs and exploring solutions.

* Many generative AIs for generic 3D assets have appeared on the market since 2024, but the results have been generally inadequate for the production scenarios I've encountered in my professional experience. The 3D mesh of the stylized brain at the art cover of this section, however, was created with one of these tools.

• WEB DEVELOPMENT

When starting to create my latest portfolio and looking for solutions, I quickly realized services such as Artstation wouldn't satisfy my needs. To properly showcase my skills, I decided to create my own portfolio from scratch and host it at fabioorsi.github.io.

With knowledge acquired in my most recent employment and heavily implementing on top of **Google's <model-viewer>**, I was able to create an **interactive** portfolio with **fully-featured 3D viewers** and **integrated 3D models**, performing well even on **mobile devices**.

3D Portfolio Preview

* Renders were automatically generated using Javascript and the real-time web viewers on fabioorsi.github.io

FURNITURE

TRANSPORTATION

3D Portfolio Preview

* Renders were automatically generated using Javascript and the real-time web viewers on fabioorsi.github.io

• SPORTS

• FURNISHINGS

• ELECTRONICS

3D Portfolio Preview

* Renders were automatically generated using Javascript and the real-time web viewers on fabioorsi.github.io

- APPAREL

- MISCELLANEOUS

Fabio Orsi osb.fabio@gmail.com fabioorsi.github.io